Composition Without Collapse

Pre-Registered Evidence for Safe but Modest Prompt-Level Multi-Adapter LoRA Composition on Apple Silicon

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Abstract

This paper studies whether multiple domain-specific LoRA adapters can be composed at prompt time on a small edge-deployed language model without retraining. The system evaluates a 4-bit Qwen2.5-1.5B-Instruct base model in MLX on Apple Silicon Unified Memory Architecture (UMA), together with 20 rank-8 domain adapters dynamically loaded into the live inference path. The central question is: when a query spans multiple domains, can routing to two adapters outperform routing to the single best adapter?

Across the core study and follow-up ablations, the work contains 1,200 real MLX inferences. Phase 1 evaluates 100 single-domain questions with four methods. The proposed AdaptiveClamp method does not beat the single-adapter baseline on semantic similarity (0.6106 vs 0.6223, delta −0.0117, 95% CI [−0.034, 0.011]). Benchmark inspection reveals the benchmark collapses to 18 question templates and 4 answer templates—a synthetic recall problem rather than a compositional reasoning test.

Phase 2 corrects this with 40 genuinely multi-domain questions across all 20 domains and 39 unique domain pairs. Under oracle routing, composition becomes directionally positive: 0.6505 vs 0.6334 for AdaptiveClamp-v2 over SingleAdapter (+0.0171, 95% CI [−0.0019, +0.0361]). This sign reversal is the central empirical finding. The improvement is real but modest, remaining below the pre-registered threshold of +0.03. Follow-up ablations reveal that a per-layer norm-ratio clamp is functionally equivalent to the simpler weight cap, while a real top-2 router recovers only ~26% of oracle headroom. The most defensible conclusion is that prompt-level multi-adapter composition is safe, occasionally beneficial, and limited by adapter geometry and base-model capacity rather than clamping mechanics.

Keywords: *LoRA composition, multi-adapter inference, Apple Silicon, on-device LLM, mixture of experts, prompt-level routing, MLX, edge AI*

At a Glance

1,200 Real MLX Inferences	**20** Domain Experts	**M3 Max** Apple Silicon / UMA	**+0.0171** MD Sim Gain (oracle)

1 Introduction

LoRA adapters make domain specialization cheap. They also surface an obvious systems question: if one adapter can specialize a base model for law, medicine, or mathematics, can two adapters be composed at inference time when a prompt genuinely spans two domains? This is especially attractive on Apple Silicon, where Unified Memory Architecture allows multiple adapter tensors to coexist without the host-to-device transfers that complicate traditional GPU serving.

I investigate this question under a stricter constraint than most mixture-of-experts work: prompt-level routing, not token-level gating. Token-level mixture-of-LoRA methods are more expressive, but they demand more routing machinery and more frequent expert recomputation. Prompt-level routing is cruder, but far easier to deploy on a small on-device system—the regime where edge AI actually lives.

The experimental path matters as much as the headline metrics. Early work framed the project as a clean negative result on synthetic single-domain questions. Later experiments added genuinely multi-domain data, clamp ablations, and routing-gap analysis. This paper tells the full story in three acts:

- On a benchmark that does not truly require composition, multi-adapter routing underperforms the simpler single-adapter baseline.
- On a benchmark that does require composition, the sign flips: two correctly chosen adapters help on average.
- The improvement is real but modest, below the pre-registered threshold for a practically meaningful win.
- Better routing helps, but only up to a point—the limiting factor appears to be representational compatibility among adapters inside the 1.5B base model.

This is a worthwhile result. It narrows the problem, identifies better next steps, and converts an initially confusing collection of experiments into a coherent empirical story.

2 System Design

2.1 Base Model and Hardware

Component	Specification
Base model	mlx-community/Qwen2.5-1.5B-Instruct-4bit
Runtime	MLX / mlx_lm
Hardware	Apple M3 Max, Unified Memory Architecture
Quantization	4-bit (4-bit weights, on-device dequant)
Total inferences	**1,200 real MLX inferences across v1, v2, v2b, v2c**

Table 0. Hardware and model configuration.

2.2 Expert Adapter Pool

The system maintains a pool of **20 domain-specific LoRA adapters**, each approximately 20.15 MB at rank-8. Domains cover a deliberately diverse range to test representational orthogonality:

Legal Analysis • Medical Diagnosis • Python Logic • Mathematics • MLX Kernels • LaTeX Formatting • Sanskrit Linguistics • Archaic English • Quantum Chemistry • Organic Synthesis • Astrophysics • Maritime Law • Renaissance Art • Cryptography • Ancient History • Music Theory • Robotics • Climate Science • Philosophy • Behavioral Economics

All adapters are dynamically loaded into the live inference path at query time. This design is only feasible at scale on Unified Memory Architecture, where GPU and CPU tensors share a flat address space and zero-copy adapter injection is achievable without host-to-device transfer overhead.

2.3 Injection Path

The live runtime (`backend/dynamic_mlx_inference.py`) does not inject adapters into every linear layer. It wraps only transformer submodules whose names contain `q_proj` or `v_proj`. This is a deliberate engineering constraint: the method is a routed composition applied to a subset of attention projections, not a full-model additive LoRA composition scheme. This scoping limits overhead while targeting the most semantically meaningful parameters.

2.4 Routing

The default `Orchestrator` is a chain-of-thought heuristic router (`backend/orchestrator.py`). In its primary path it returns a one-hot top-1 routing vector. This detail is important: for the multi-domain evaluation in Phase 2, oracle routing is used for K=2 methods so the experiment measures compositional value independently of top-2 discovery quality.

2.5 Clamping Formulations

Two clamping strategies are studied. The simpler runtime weight cap, used in the primary v1/v2 experiments:

$$\hat{m} = \Sigma_i \min(w_i, c) \cdot s_i(x)$$

The follow-up norm-ratio clamp (v2b ablation), applied per-layer:

$$m_\square = \Sigma_i w_i \cdot s_{i\square}(x), \quad \gamma_\square = \min(1, c \cdot \|z_\square\|_2 / (\|m_\square\|_2 + \varepsilon)), \quad \hat{z}_\square = z_\square + \gamma_\square m_\square$$

The norm-ratio clamp is the theoretically cleaner formulation: it adapts the scaling factor per layer based on the ratio of base activation to adapter contribution, preventing outsized adapter influence in sensitive layers. The ablation study (Section 5.5) tests whether this advantage materializes empirically.

3 Experimental Design

3.1 Metrics

Three primary metrics are reported throughout:

- Semantic Similarity: cosine similarity between generated text and the reference answer using sentence-transformers/all-MiniLM-L6-v2. This is the primary quality proxy.
- Perplexity: log-likelihood of the reference answer under the current routing configuration. Lower is better.
- Latency: wall-clock generation time in seconds. Overhead is computed relative to the SingleAdapter baseline.

> *The semantic similarity metric is useful for within-system comparisons but is an embedding-based proxy, not a factual correctness measure. On templated benchmarks, lexical drift in the generated output can dominate the score independently of actual answer quality.*

3.2 Phase 1: Single-Domain Benchmark

Phase 1 evaluates 100 synthetic domain questions (20 domains × 5 questions each) under four routing methods. This phase contributes 400 real inferences.

Method	K	Clamp	Routing
Baseline	0	0.001	None
SingleAdapter	1	0.5	CoT top-1
UnclampedMix	2	999	CoT top-2
AdaptiveClamp	**2**	**0.5**	**CoT top-2**

Table 1. Phase 1 method configurations.

3.3 Phase 2: Mixed Single-Domain and Multi-Domain Evaluation

Phase 2 expands the evaluation with 40 new multi-domain questions designed to genuinely require compositional reasoning. The multi-domain benchmark has:

- 40 questions in total
- All 20 domains appearing at least once
- 39 unique unordered domain pairs
- One repeated pair: PYTHON_LOGIC × ROBOTICS (for consistency testing)

Method	K	Clamp	Routing

Baseline	0	0.001	None
SingleAdapter	1	0.5	CoT top-1
AdaptiveClamp-v2	**2**	**0.5**	**Oracle**
UnclampedMix-v2	2	999	Oracle

Table 2. Phase 2 method configurations. Oracle routing uses ground-truth domain labels for K=2 selection.

3.4 Follow-up Ablations

Two additional studies operate on the 40-question multi-domain split, each contributing 120 real inferences:

- v2b — Clamp Ablation: weight-cap vs norm-ratio clamp, holding routing constant at oracle K=2.
- v2c — Routing-Gap Ablation: oracle K=2 vs real top-2 router, holding clamping constant at norm-ratio.

3.5 Pre-Registered Thresholds

All primary hypotheses were registered before running the experiments. Results are evaluated against these thresholds without post-hoc adjustment.

Phase	Hypothesis	Threshold	Outcome
v1	Delta_SIM(AdaptiveClamp - SingleAdapter)	> +0.05	**FAIL**
v1	PPL(AdaptiveClamp) <= PPL(SingleAdapter)	pass/fail	**PASS**
v1	Latency overhead	<= 10%	**PASS**
v2 SD	Non-inferiority vs SingleAdapter	>= -0.005	**PASS**
v2 MD	Delta_SIM(AdaptiveClamp-v2 - SingleAdapter)	> +0.03	**NEAR**
v2	Perplexity preservation	pass/fail	**PASS**
v2	Latency overhead	<= 15%	**PASS**

Table 3. Pre-registered hypotheses and outcomes. NEAR indicates the result approached but did not clear the threshold.

4 Benchmark Diagnosis: A Necessary Detour

The strongest contribution of this work is not numerical but methodological. After normalizing domain names, the 100-question single-domain benchmark collapses into:

- 18 normalized question templates
- 4 normalized answer templates
- 58 instances of one "fundamental theorem of {domain}" answer pattern
- 40 instances of one "orthogonal projections in {domain}" answer pattern
- Only 2 genuinely unique answers: res ipsa loquitur (legal) and a BFS code snippet

> *The Phase 1 benchmark is primarily a test of synthetic domain-token recall, not compositional reasoning. If the correct answer depends on exactly one domain and a second adapter is almost always redundant, failure of K=2 composition is the expected null result—not evidence against the method.*

This diagnosis is what makes the Phase 2 sign reversal interpretable rather than contradictory. It also illustrates a general hazard in adapter composition research: evaluation benchmarks that appear domain-diverse can collapse to a small set of template patterns that reward single-adapter recall.

The fix in Phase 2 is straightforward: construct questions that require both domain vocabularies to appear in the reference answer. The 40-question multi-domain benchmark was built with this criterion enforced by manual inspection of each item.

5 Results

5.1 Phase 1: Multi-Adapter Composition Loses on Single-Domain Recall

Method	Avg Sim	Avg PPL	Avg Latency (s)
Baseline	0.6196	64.46	2.803
SingleAdapter ★	**0.6223**	60.89	2.695
AdaptiveClamp	0.6106	58.03	2.673
UnclampedMix	0.5573	**51.21 ★**	**2.511 ★**

Table 4. Phase 1 results. ★marks best value per column. SingleAdapter wins on semantic similarity; UnclampedMix achieves lowest PPL but at the cost of coherence.

The pre-registered primary comparison fails:

- Delta_SIM(AdaptiveClamp − SingleAdapter) = −0.0117
- 95% CI: [−0.034, 0.011]
- Prompt-level wins / losses / ties: 45 / 52 / 3

Two patterns stand out. First, the simpler single-adapter baseline outperforms on average, though AdaptiveClamp does win in some domains (MATHEMATICS: +0.0443, MEDICAL_DIAGNOSIS: +0.0295) while losing badly in others (MARITIME_LAW: −0.1449). Second, unclamped mixing is clearly unsafe: UnclampedMix yields semantic similarity below 0.1 on 3 prompts and below 0.2 on 7 prompts, with a minimum of 0.0312.

> *Result: Naive additive mixing is dangerous. Bounded mixing is much safer. Safe mixing still does not beat single-adapter routing when the benchmark does not require composition.*

5.2 Phase 2 Sanity Split: Single-Domain Performance Preserved

Method	Avg Sim	Avg PPL	Avg Latency (s)
Baseline	0.6090	64.5	3.700
SingleAdapter ★	**0.6064**	60.9	**3.571 ★**
AdaptiveClamp-v2	0.6058	57.9	3.657
UnclampedMix-v2	0.6041	**52.3 ★**	3.623

Table 5. Phase 2 sanity split (100 single-domain questions re-evaluated in v2 harness). AdaptiveClamp-v2 is non-inferior to SingleAdapter.

The non-inferiority hypothesis passes cleanly:

- Delta_SIM(AdaptiveClamp-v2 − SingleAdapter) = −0.0006

- 95% CI: [−0.0076, 0.0064]
- Prompt-level wins / losses / ties: 29 / 29 / 42

This is precisely the desired behavior for a production system: composition no longer imposes a meaningful penalty on single-domain queries once routing is appropriately constrained.

5.3 Phase 2 Main Result: Composition Helps on Genuine Multi-Domain Prompts

Method	Avg Sim	Avg PPL	Avg Latency (s)
Baseline	0.6473	12.7	4.059
SingleAdapter	0.6334	12.7	**4.057 ★**
AdaptiveClamp-v2 ★	**0.6505 ★**	**12.6 ★**	4.090
UnclampedMix-v2	0.6505	12.6	4.100

Table 6. Phase 2 multi-domain results. This is the paper's primary empirical finding: the sign reverses under a properly matched benchmark.

This is the most important empirical result:

- Delta_SIM(AdaptiveClamp-v2 − SingleAdapter) = +0.0171
- 95% CI: [−0.0019, +0.0361]
- Prompt-level wins / losses / ties: 28 / 11 / 1

> *The sign has flipped relative to Phase 1. Once the benchmark genuinely requires two domains and the K=2 method is routed to the correct pair, composition is beneficial on average. The win is real but modest: it misses the pre-registered +0.03 threshold and sits only +0.0032 above the no-adapter baseline.*

5.4 Gains Are Highly Pair-Dependent

The multi-domain average conceals large per-pair heterogeneity. Selected items from the top and bottom of the distribution:

Item	Domain Pair	Delta vs SingleAdapter
md_32	**MEDICAL_DIAGNOSIS x MATHEMATICS**	**+0.3029**
md_19	LATEX_FORMATTING x MATHEMATICS	+0.1096
md_20	LEGAL_ANALYSIS x CRYPTOGRAPHY	+0.0824
md_02	MARITIME_LAW x BEHAVIORAL_ECONOMICS	+0.0671

md_34	SANSKRIT_LINGUISTICS x ANCIENT_HISTORY	+0.0644
md_09	PYTHON_LOGIC x ROBOTICS	**-0.1252**
md_25	CLIMATE_SCIENCE x ORGANIC_SYNTHESIS	-0.0610
md_38	ARCHAIC_ENGLISH x LEGAL_ANALYSIS	-0.0293

Table 7. Best and worst-performing domain pairs. Green = composition helps; red = composition hurts. The repeated PYTHON_LOGIC × ROBOTICS failure is notable.

This pattern is consistent with an adapter-geometry hypothesis: domain pairs occupying sufficiently different representational subspaces compose productively, while pairs sharing overlapping machinery (e.g., two procedural-reasoning domains) interfere. The repeated failure on PYTHON_LOGIC × ROBOTICS in both md_09 and md_24 is the clearest example.

5.5 Clamp Ablation: The Cleaner Formulation Does Not Move the Needle

Method	Clamp	Avg Sim	Avg PPL	Latency (s)
SingleAdapter	weight_cap	0.6334	12.7	4.008
AC-v2-WeightCap	weight_cap	0.6505	12.6	4.055
AC-v2-NormRatio	norm_ratio	0.6502	12.6	4.221

Table 8. Clamp formulation ablation on the 40-question multi-domain split. Weight-cap and norm-ratio are nearly indistinguishable.

Key deltas:

- Delta_SIM(WeightCap − SingleAdapter) = +0.0171
- Delta_SIM(NormRatio − SingleAdapter) = +0.0168
- Delta_SIM(NormRatio − WeightCap) = −0.0003

The theoretically cleaner per-layer norm-ratio clamp is essentially equivalent to the simpler runtime weight cap. The most plausible explanation: the raw adapter contribution is small enough relative to the base activation that $\gamma_{\square} \approx 1.0$ almost everywhere, making the two formulations functionally identical on this model family.

5.6 Routing-Gap Ablation: Better Routing Helps, But the Oracle Ceiling Is Already Low

Method	Routing	Avg Sim	Avg PPL	Lat (s)	Avg K
SingleAdapter	CoT top-1	0.6296	12.8	4.18	1.00

AC-v2-RealRouter	CoT top-2	0.6350	12.7	4.17	1.75
AC-v2-Oracle ★	**Oracle top-2**	**0.6502 ★**	**12.6 ★**	4.21	**2.00**

Table 9. Routing-gap ablation. Oracle routing represents the upper bound for any routing improvement.

The bottleneck ranking is clear:

- Oracle headroom over SingleAdapter: +0.0206
- Realized gain with a real top-2 router: +0.0054
- Fraction of oracle headroom recovered: ~26%

> *Conclusion: Composition capacity is limited even under oracle routing. Routing quality matters, but improving it alone cannot create a large effect that is not already there. Clamp formulation is the smallest factor among the three.*

6 Interpreting the Full Experimental Arc

6.1 The Phase 1 Negative Is Real, but Narrow

Phase 1 is a genuine negative result on single-domain synthetic recall. On that task, the extra adapter acts as noise, and the result correctly reflects that. The limitation is that this is a conclusion about evaluation design as much as about the method.

6.2 The Phase 2 Sign Reversal Is Also Real

A mean effect that changes sign from −0.0117 to +0.0171 across 40 items is too large a shift to dismiss as noise or cherry-picking. The system learned something important: the original negative conclusion was partly an artifact of the evaluation.

6.3 The Ceiling Is Low at 1.5B Scale

Under oracle K=2, the system reaches 0.6505 on the multi-domain split—barely above the no-adapter baseline of 0.6473. If perfect routing yields only two similarity points over the single-adapter baseline, the main constraint is unlikely to be router engineering. Adapter orthogonality and base-model capacity appear to be the deeper bottlenecks.

6.4 Perplexity Repeatedly Improves Without Corresponding Semantic Gains

Multi-adapter injection consistently lowers perplexity even when it does not improve semantic similarity. The interpretation is that composition can make the model locally more confident in the style or token distribution of reference answers without reliably improving answer selection. On templated synthetic data, this discrepancy becomes even easier to produce.

7 Limitations

The following limitations are surfaced in the main text, not relegated to footnotes.

7.1 Small and Synthetic Benchmarks

The single-domain benchmark is heavily templated, and the multi-domain benchmark contains only 40 questions. Results on the multi-domain split carry meaningful uncertainty; confidence intervals reflect this throughout.

7.2 Metric Sensitivity

The main quality metric is embedding cosine similarity to a single reference answer. This is a pragmatic choice but is not equivalent to factual correctness or usefulness. A task that produces fluent but factually wrong text can achieve high similarity scores if the error is lexically close to the reference.

7.3 Partial-Model Injection

The live composition path only modifies `q_proj` and `v_proj`. Any general claim about "multi-adapter composition" should be understood as restricted to this attention-projection scope.

7.4 Router Mismatch Across Experiments

The experiments span one-hot CoT routing, a real top-2 extension, embedding and classifier routers, and a gated router. These are informative but not all directly comparable. Only the CoT and oracle routing experiments are tightly tied to the pre-registered framing.

7.5 Exploratory Artifacts Are Secondary

Files such as `FINAL_EXPERIMENT_REPORT.md` and `results/mistral_vs_synapta_verified.md` contain interesting later-stage claims including autonomous routing and Mistral comparisons. These are useful engineering evidence but sit outside the cleanest pre-registered comparisons and should be treated as exploratory follow-up, not primary proof.

8 Practical Guidance for Practitioners

For engineers and researchers building on this work:

1	Do not use unclamped multi-adapter mixing in production. Unclamped injection produces semantic similarity below 0.1 on a meaningful fraction of prompts.
2	Expect K=2 composition to underperform on single-domain queries. Routing should be able to suppress the second adapter when only one domain is relevant.
3	For genuinely cross-domain queries, measurable headroom exists for composition—but at 1.5B scale it is small. Set expectations accordingly.
4	Prioritize adapter compatibility and higher-capacity base models before investing in more elaborate clamp formulations. The norm-ratio clamp provides no measurable gain over a simple weight cap.
5	Treat routing as a meaningful but secondary bottleneck. A real top-2 router recovers only ~26% of oracle headroom; engineering effort is better spent on adapter training strategies that improve orthogonality.

9 Conclusion

This paper does not show that prompt-level multi-adapter LoRA composition is a dead end. It shows something more specific and more useful.

On a synthetic single-domain benchmark, K=2 composition fails to beat a single routed adapter. That negative result is genuine. But once the evaluation is redesigned around genuinely multi-domain questions, the effect reverses sign: correctly routed composition is beneficial on average. The catch is that the benefit is modest and remains below the pre-registered threshold for a practically meaningful win.

The most defensible conclusion: composition can be made safe; composition can help when the task truly requires it; composition is not yet strong enough at 1.5B scale to justify broad claims of decisive advantage over single-expert routing.

That is a worthwhile result. It narrows the problem, identifies better next steps—adapter geometry, base-model scale, orthogonality-aware training—and converts an initially confusing collection of experiments into a coherent empirical story. The clearest path forward is not more elaborate clamping, but better-engineered adapters and larger base models.

Appendix A: Reproducibility

All experiments are reproducible from the repository. Core entry points:

```
# Phase 1: single-domain benchmark
python3 src/eval/real_benchmark.py

# Phase 2: mixed SD + MD benchmark
python3 src/eval/run_eval_v2.py --real --split both

# Phase 2b: clamp ablation
PYTHONUNBUFFERED=1 python3 src/eval/run_eval_v2b.py --phase clamp --real

# Phase 2c: routing-gap ablation
PYTHONUNBUFFERED=1 python3 src/eval/run_eval_v2b.py --phase routing --
real
```

Primary raw result artifacts:

- results_db.jsonl
- results/real_benchmark_results.json
- results/v2_both_raw.jsonl
- results/v2_md_clamp_ablation.jsonl

- results/v2_md_routing_ablation.jsonl

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